

Unit 1: Molecules, Diet, Transport and Health

IAS compulsory unit

Externally assessed

Unit description

Introduction

This unit includes a consideration of molecules that are important in biology – including water, carbohydrates, lipids, proteins and nucleic acids, providing a basis for many areas of biology. This leads to the relevance of diet to health and the cardiovascular system in particular. The unit includes cell membrane transport processes, such as diffusion and active transport, proteins, enzymes and protein synthesis. This unit also includes an understanding of the genetic code and how mutations can result in disorders. Students will also consider techniques for genetic screening and the associated ethical and social issues.

Practical skills

In order to develop their practical skills, students should be encouraged to carry out a range of core practical experiments related to this topic. There are also opportunities to carry out additional practical work, such as investigating the structure of a mammalian heart.

Mathematical skills

There are opportunities for the development of mathematical skills in this unit, including tabulation and graphical treatment of data, concentrations and using appropriate units for physical quantities, calculation of percentage changes, and finding the initial rate of enzyme-catalysed reactions. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

Assessment information

- First assessment: January 2019.
 - The assessment is 1 hour and 30 minutes.
 - The assessment is out of 80 marks.
 - Candidates must answer all questions.
 - The paper may include multiple-choice, short-open, open-response, calculations and extended-writing questions.
 - The paper will include a minimum of 8 marks that target mathematics at Level 2 or above.
 - Calculators may be used in the examination. Please see *Appendix 8: Use of calculators*.
 - Candidates will be expected to apply their knowledge and understanding to familiar and unfamiliar contexts.
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Topic 1 – Molecules, Transport and Health

Candidates will be assessed on their ability to:

1.1	understand the importance of water as a solvent in transport, including its dipole nature
1.2	(i) know the difference between monosaccharides, disaccharides and polysaccharides, including glycogen and starch (amylose and amylopectin) (ii) be able to relate the structures of monosaccharides, disaccharides and polysaccharides to their roles in providing and storing energy <i>β-glucose and cellulose are not required in this topic.</i>
1.3	CORE PRACTICAL 1 Use a semi-quantitative method with Benedict's reagent to estimate the concentrations of reducing sugars and with iodine solution to estimate the concentrations of starch, using colour standards.
1.4	know how monosaccharides (glucose, fructose and galactose) join together to form disaccharides (maltose, sucrose and lactose) and polysaccharides (glycogen, amylose and amylopectin) through condensation reactions forming glycosidic bonds, and how these can be split through hydrolysis reactions
1.5	(i) know how a triglyceride is synthesised by the formation of ester bonds during condensation reactions between glycerol and three fatty acids (ii) know the differences between saturated and unsaturated lipids
1.6	understand why many animals have a heart and circulation (mass transport to overcome the limitations of diffusion in meeting the requirements of organisms)
1.7	understand how the structures of blood vessels (capillaries, arteries and veins) relate to their functions
1.8	know the cardiac cycle (atrial systole, ventricular systole and cardiac diastole) and relate the structure and operation of the mammalian heart, including the major blood vessels, to its function <i>Details of myogenic stimulation are not needed at IAS.</i>

RECOMMENDED ADDITIONAL PRACTICAL	
Investigate the structure of a mammalian heart by dissection.	
1.9	(i) understand the role of haemoglobin in the transport of oxygen and carbon dioxide (ii) understand the oxygen dissociation curve of haemoglobin, the Bohr effect and the significance of the oxygen affinity of fetal haemoglobin compared with adult haemoglobin
1.10	understand the course of events that leads to atherosclerosis (endothelial dysfunction, inflammatory response, plaque formation, raised blood pressure)
1.11	understand the blood clotting process (thromboplastin release, conversion of prothrombin to thrombin and fibrinogen to fibrin) and its role in cardiovascular disease (CVD)
1.12	know how factors such as genetics, diet, age, gender, high blood pressure, smoking and inactivity increase the risk of cardiovascular disease (CVD)
1.13	understand the link between dietary antioxidants and the risk of cardiovascular disease (CVD)
1.14	CORE PRACTICAL 2 Investigate the vitamin C content of food and drink.
1.15	be able to analyse and interpret quantitative data on illness and mortality rates to determine health risks, including distinguishing between correlation and causation and recognising conflicting evidence
1.16	be able to evaluate the design of studies used to determine health risk factors, including sample selection and sample size used to collect data that is both valid and reliable
1.17	understand why people's perception of risks are often different from the actual risks, including underestimating and overestimating the risks due to diet and other lifestyle factors in the development of heart disease
1.18	(i) be able to analyse data on the possible significance for health of blood cholesterol levels and levels of high-density lipoproteins (HDLs) and low-density lipoproteins (LDLs) (ii) know the evidence for a causal relationship between blood cholesterol levels (total cholesterol and LDL cholesterol) and cardiovascular disease (CVD)
1.19	understand how people use scientific knowledge about the effect of diet, including obesity indicators, such as body mass index and waist-to-hip ratio, exercise and smoking to reduce their risk of coronary heart disease
1.20	know the benefits and risks of treatments for cardiovascular disease (CVD) (antihypertensives, statins, anticoagulants and platelet inhibitors)